RAFAEL M. CHAVEZ

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Education

Georgia Institute of Technology, Atlanta, GA Aug. 2023 – May 2027 (Expected)
B.S. Mechanical Engineering, Minor in Applications of AI/Machine Learning *GPA: 3.9 / 4.0*

Work Experience

Autonomy and Controls Engineering Intern May 2026 – Nov. 2026
Sea Machines Robotics

- Tested and evaluated radar sensor hardware to ensure reliable performance for autonomous vessel deployments.
- Built a streamlined data ingestion application supporting live mission data capture for downstream analysis.
- Developed a vessel-agnostic dynamic model parameter estimation tool achieving sub-5% average accuracy in simulation, supporting collision avoidance, fuel optimization, and trajectory planning.
- Applied digital signal filtering techniques to improve fidelity of parameter state estimation model.
- Validated estimated hydrodynamic parameters using motion model simulations with injected Markov-Gauss sensor noise and physical environmental wave disturbances.

Process Engineering Intern May 2025 – Aug. 2025
Amazon Robotics

- Developed a modular Shadow Board fabrication, infrastructure, and procurement system to promote 5S principles, creating a tooling database of over 150 discrete tools.
- Streamlined Shadow Board design and procurement into a robust, high-ceiling process, reducing requisite labor by 94%.
- Conducted cycle time studies and developed SOPs defining discrete scopes of work for material line replenishment and warehouse storage re-organization across manufacturing operations.
- Cut part delivery time 81%, increased discrete materials delivered per cycle 18x, and reduced station headcount 60%.
- Drove an 85% increase in technician task efficiency through Pareto analysis, and trained manufacturing technicians on best-practice SOPs.
- Identified bottlenecks via time cycle analysis to support labor/headcount rebalancing, cutting manufacturing downtime and reducing headcount by 22% through root cause analysis.

Research Experience

Project Lead and Research Assistant Dec. 2025 – Nov. 2026
Institute for Robotics and Intelligent Machines (IRIM), Georgia Tech

- Deployed GPT-4o and OpenCV to detect, locate, and manipulate objects with UR5 and Rainbow robot arms.
- Trained an RF-DETR detection model to improve end-effector relative object manipulation and orientation estimation.
- Built a $50,000 cost-saving machine vision part inspection prototype to compete in the TE Connectivity AI Cup.

Research Assistant Aug. 2024 – May 2025
Exoskeleton & Prosthetic Intelligent Controls (EPIC) Lab, Georgia Tech

- Engineered electrical junction box hardware for a hip exoskeleton assistive device targeting stroke and cerebral palsy patient rehabilitation, as part of the Mechatronics and Hardware Development sub-team.
- Analyzed exoskeleton biomechanics via VICON motion capture and simulation software to inform design decisions.
- Cut hardware housing fixture mass 55% and volume 32% while optimizing cable routing and component cooling.
- Designed parts in SolidWorks and manufactured using 3D printing and pre-impregnated carbon fiber layering.

Research Assistant Jan. 2024 – May 2024
Chip Scale Power & Energy Project, Georgia Tech Research Institute

- Fabricated electrochemical double layer (ECDL) supercapacitor chips and electrodes using carbon nanotubes (CNTs) embedded within a silicon wafer.
- Functionalized CNTs via nanomaterial synthesis, cleanroom microfabrication, and atomic layer deposition.
- Hermetically packaged chip-scale devices for incorporation as viable prototypes on mission architecture.
- Characterized electrodes and analyzed process efficiency of atomic layer deposition and plasma-enhanced chemical vapor deposition through data analysis.

Publications and Awards

- **R. Chavez**, "Roboflow Detection Transformer for Manufacturing Defect Inspection with an application to Catheter Shaft Inspection," *IEOM 2026 World Congress*, Irvine, CA, Oct. 2026.
  1st Place, Best Undergraduate Research Paper Award Track Sponsored by Siemens
- **Eagle Scout**, Troop 1 Boxborough, Boy Scouts of America (May 2023)

Projects

RoBobby Fischer Aug. 2026 – Present

Autonomous Chess-Playing Robot Mechatronics Project (Personal Project)

- Designing and building a chess-playing robot arm from scratch, including custom 3D-printed linkages and hardware architecture.
- Selecting and integrating motors, sensors, and microcontrollers (Arduino, Raspberry Pi) to drive arm actuation and sensing.
- Developing a computer vision model to detect board state and piece positions in real time.
- Integrating a Stockfish API to compute optimal chess moves and translate engine output into physical arm motion through a control stack built from the ground up.

Mobile Robot Navigation Jan. 2026 – May 2026

Robotics Coursework, Georgia Institute of Technology

- Built forward/inverse kinematic models in MATLAB for serial and parallel manipulators, optimizing trajectory planning and closed-loop end-effector control.
- Programmed TurtleBot3 autonomous navigation in ROS, fusing LiDAR, IMU, and odometry data for obstacle avoidance and path following.
- Trained CNNs for robotic object classification, tuning architectures and hyperparameters to maximize accuracy.

Leadership and Organizations

President Sept. 2026 – Present

Industrial Engineering and Operations Management Society GT Chapter

- Founded and chartered the chapter as a Georgia Tech Registered Student Organization and a chapter of IEOM Society International, and drafted its constitution.
- Chair the executive board and guide the organization toward growth.
- Defined the chapter's mission to weight academic and industry paths equally, covering research opportunities, undergraduate research fairs, conference and paper-submission coaching, industry networking, and projects with industry partners.

Chairman, Judicial Board & Membership Well-Being Nov. 2025 – Present

Sigma Phi Epsilon

- Elected chairman of the judicial board and liaison to the executive board within the organization.
- Led over 25 disciplinary hearings on topics ranging from finances to academics and behavioral matters.
- Increased operational efficiency of judicial proceedings by standardizing hearing procedures and documentation.
- Improved mental health and risk management resources to increase membership involvement and well-being.

Director of Corporate Relations Jul. 2025 – May 2026

Pi Tau Sigma Mechanical Engineering Honor Society, Nu Chapter

- Served on the executive board of the GT chapter of the Mechanical Engineering Honor Society.
- Organized recruiting events and fostered professional relationships between recruiters and prospective students.

Student Advisory Board Member Mar. 2025 – May 2026

Woodruff School of Mechanical Engineering, Georgia Tech

- Served as a student representative to the Woodruff School of Mechanical Engineering.
- Advised college leadership on methods to improve student life by addressing issues pertinent to the student body.

Undergraduate Teaching Assistant and Mentor Jan. 2024 – Dec. 2024

GT Tutoring and Academic Support Center

- Organized and conducted orientation for incoming Teaching Assistants, ensuring a smooth onboarding process, and facilitated biweekly mentorship check-ins for the TA team.
- Facilitated active learning and collaboration for 50+ students in Statics/Mechanics lectures, office hours, and recitation sessions.
- Collaborated with faculty stakeholders to create student-centered resources as part of the mechanical engineering curriculum.

Member | *RoboRacing, Georgia Tech Competitive Robotics Club* Sept. 2026 – Present

Member | *SHPE, ASME, IEEE* Aug. 2023 – Present

Member | *Industrial Engineering and Operations Management (IEOM) Society International* June 2026 – Present

Technical Skills

Programming & Software: MATLAB, Python, C, C++, Java, MOOS-IvP, Excel, Linux, WSL, Git
Robotics & AI: ROS, PID Controller Tuning, OpenCV, YOLO, RF-DETR, Deep Learning, CNNs, Reinforcement Learning, Neo4j, Cypher, RoboFlow
Embedded Systems: STM32 (GPIO, ADC, UART), Arduino, Raspberry Pi
CAD & Design: SolidWorks, Autodesk Fusion 360
Manufacturing & Methods: 3D Printing, Machining, Lathe, Soldering, Carbon Fiber Layering, Rapid Prototyping, Six Sigma, 5S
Relevant Coursework: Robotics, Mechatronics, Machine Learning, Probability & Statistics, System Dynamics, Circuit Design, Machine Design, Dynamics, Statics, Heat Transfer, Fluid Mechanics, Thermodynamics, Mechanics of Materials